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Thermodynamic assessment of the Ho–Ga system

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Abstract
By means of the CALPHAD (CALculation of PHAse Diagrams) technique, the holmium–gallium binary system was critically assessed. The thermodynamic parameters involved in the excess term of the Gibbs energy phases are optimized based on all the experimental data available in the literature. The Ho–Ga system contains five intermetallic compounds: $HoGa_6$, $HoGa_3$, $HoGa_2$, HoGa and Ho_5Ga_3, which are all treated as stoichiometric phases. The term ^{exe}G of the Gibbs energy of the liquid phase is described, using the Redlich–Kister, Kaptay equations and the association solution model. The primary phases αHo (Hcp) and Ga (orthorhombic) are treated as pure stoichiometric phases. The phase diagram and model parameters were derived from a thermodynamic optimization. More experimental work on this system may be needed to confirm our results.

Keywords Ho–Ga system · Phase diagram · Thermodynamic evaluation · CALPHAD technique · Redlich–Kister · Kaptay equations and association solution model

Introduction

First, the gallium element has many applications; it is used in the form of arsenide (GaAs) and phosphide (Gap) [1] as substrates in integrated digital circuits. Moreover, the transfer of electrons by gallium is three times faster than that obtained with silicon-based semiconductors. In addition, gallium-based semiconductors are highly radiation resistant and can be used to convert electrical signals into optical signals. It is also used in very low-power lighting and in the development of solar energy (photovoltaic) [1].

In addition, rare earth intermetallic compounds have a very important industrial interest because of their richness of physical properties. In general, two areas of activity depend on rare earths: emerging sectors, including ceramics, permanent magnets and battery alloys; and mature sectors such as glass manufacturing, catalysts, lighting and metallurgy. Moreover, intermetallic rare earth compounds with gallium have attracted a great deal of attention for the theoretical study of the solid state and for many industrial applications.

In the current work, the Ho–Ga binary system has been optimized using the CALPHAD technique [2]. In this method, the thermodynamic properties of Ho–Ga systems are analysed using thermodynamic models for Gibbs free energy of different phases. The thermodynamic parameters involved in the models are optimized from experimental data and a phase diagram.

Experimental information

Equilibrium diagram

The holmium–gallium phase diagram was published for the first time by Yatsenko et al. [3, 4]. These authors indicate the existence of four intermetallic compounds: $HoGa_3$, $HoGa_2$, HoGa and Ho_5Ga_3. It can be deduced that the R-Ga binary systems published by the authors [5–8] that the Ho–Ga system version presented by Yatsenko et al. [4] lacked an intermetallic compound $HoGa_6$, and by interpretation all gallium-based alloys with

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rare earths are known by the existence of the R–Ga_6 (R = Ho, Nd, Pr…) phase.

Recently, Pelleg and Kimmel [9] published a work on the Ho–Ga system in the gallium-rich part, which shows that no solid solubility of gallium in $HoGa_2$ was found, in order to confirm the presence of $HoGa_3$, to determine the limit of the peritectic temperature of $HoGa_6$ and to redetermine the liquidus in the gallium content range 66.7–100 at.% Ga (Fig. 1).

Two other intermetallic compounds, Ho_3Ga_5 and Ho_3Ga_2, not shown in the phase diagram given by [4, 9], were mentioned by Yatsenko et al. [10, 11], respectively. Massalski [12] (Fig. 2) also plotted an updated version of the Ho–Ga binary system, given by [4, 9], in which the compound Ho_3Ga_5 appears. But he did not define the types

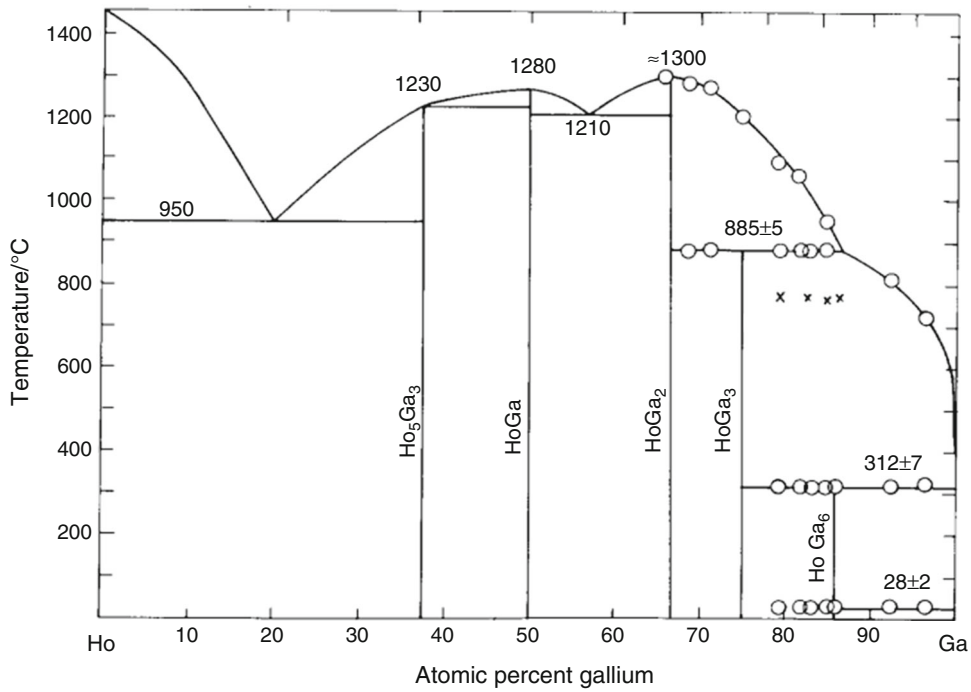

Fig. 1 The phase diagram of the Ho–Ga system reviewed by Pelleg [9] with X, unexplained DTA data for this author

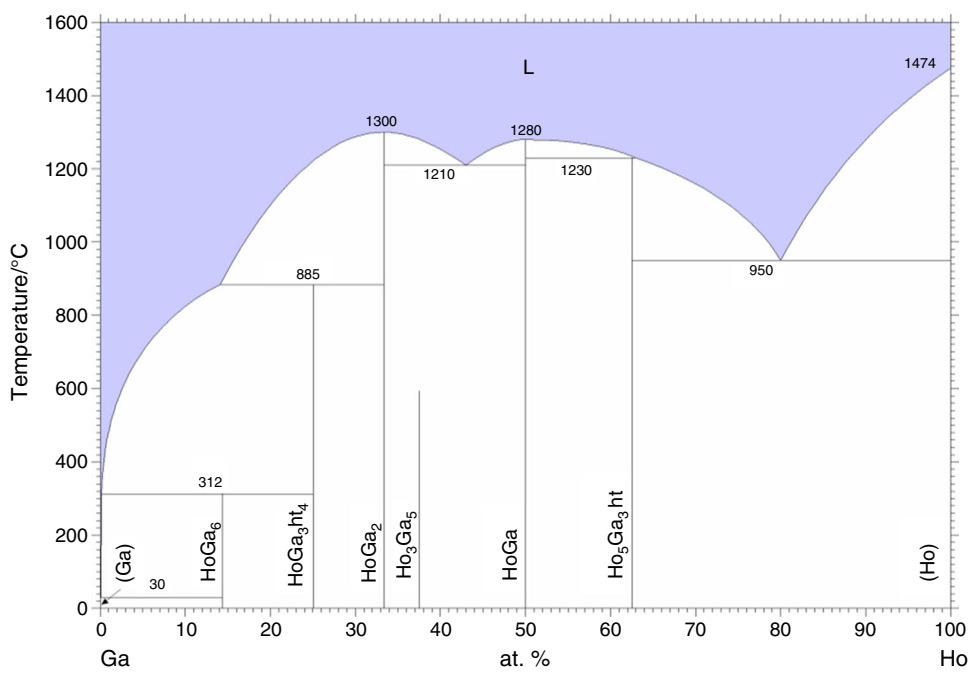

Fig. 2 The phase diagram of the Ho–Ga system reviewed by Massalski [12]

of reactions performed by this compound and related compounds. Cirafici and Franceschi [13] cited the existence of five polymorphic modifications of the binary compound $HoGa_3$ with the structure types $TiNi_3$, $HfNi_3$, $HoAl_3$,$AuCu_3$ and $PuAl_3$. However, recent and accurate studies on the binary compound Sc_5Ga_3, conducted by Corbett and co-authors [14, 15], have shown that this phase in high temperature is of type Ba_5Si_3, and at about 800 °C, a slight modification of type Ba_5Si_3 is performed. Nevertheless, no further work was done to establish the exact transformation temperature on the compound Ho_3Ga_5. In this work, we will compare our results with those given by the Pelleg and Kimmel [9] diagram (Fig. 1).

Thermodynamic data

Yamshchikov et al. [16] determined by the Emf method (at 660–1065 K) the enthalpies of formation of two intermetallic compounds $HoGa_6$ and $HoGa_3$ (Table 1). Meschel and Kleppa [17] measured the standard enthalpy of formation of $HoGa_3$ compound using the dynamic differential calorimetry method (Table 1). The other compounds are calculated by the Miedema model (Table 1) [18].

Crystallographic data

The crystallographic data for all intermediate compounds formed in the Ho–Ga system are given in Table 2.

Thermodynamic models

Thermodynamic description of pure elements

According to SGTE [24], the Gibbs free energy of a pure element i (i = Ho, Ga) in the phase φ (φ = Hcp and orthorhombic) in its standard reference (SER) state is given according to the temperature by the formula:

Table 1 Comparison between the enthalpies of formation of the intermetallic compounds of the holmium–gallium system derived from the current work and taken from the literature

Compounds	Enthalpies of formation/KJ Mol^{-1} at^{-1}	Entropies of formation/J K^{-1} mol^{-1} at^{-1}	Method used	References
Ho_5Ga_3	− 61.02	–	Miedema's model	[18]
	− 56.45	1.3493	Optimization[R−K]	Current work
	−62.21	1.3238	Optimization[K]	Current work
	−61.2	9.00	Optimization[A]	Current work
HoGa	−71.34	–	Miedema's model	[18]
	−71.00	4.7116	Optimization[R−K]	Current work
	−70.00	0.8896	Optimization[K]	Current work
	−67.6	7.330	Optimization[A]	Current work
$HoGa_2$	−62.92	–	Miedema's model	[18]
	−72.00	3.5907	Optimization[R−K]	Current work
	−71.00	1.8299	Optimization[K]	Current work
	−69.04	3.801	Optimization[A]	Current work
$HoGa_3$	−50	–	Miedema's model	[18]
	−57.8	–	Emf (660 – 1065 K)	[16]
	−56.8	–	Dir.Syn.Calo 137362 K	[17]
	−56.80	0.1403	Optimization[R−K]	Current work
	−56.80	−0.0523	Optimization[K]	Current work
	−56.80	2.6898	Optimization[A]	Current work
$HoGa_6$	−28.95	–	Miedema's model	[18]
	−33.1	–	Emf(660 – 1065 K)	[16]
	−35.00	0.8342	Optimization[R−K]	Current work
	−33.00	−2.6685	Optimization[K]	Current work
	−33.10	−1.2683	Optimization[A]	Current work

[R–K] Redlich–Kister equation

[K] Kaptay equation

[A] Association model

$$G_i^{\varphi}(T) = {}^0G_i^{SER}(T) - {}^0H_i^{SER}(298.15\ \mathrm{K}) = f(T) \quad (1)$$

$$G_i^{\varphi}(T) = a + b \times T + c \times T \times \ln(T) + d \times T^2 + e \times T^3 + f \times T^{-1} + g \times T^7 + h \times T^{-9} \quad (2)$$

where ${}^0H_i^{SER}(298.15\ \mathrm{K})$ is the molar enthalpy of the element i in its standard element reference (SER) state Hcp for holmium and orthorhombic for gallium, and T is the absolute temperature; a, b, c, …, h are the coefficients.

The Gibbs free energy of the element $i (i = \mathrm{Ga, Ho})$, in its SER state is denoted by $\mathrm{GHSER_i}$ as follows (at 298.15 K):

$$\mathrm{GHSER_{Ho}} = {}^0G_{Ho}^{Hcp_A3} = G_{Ho}^{Hcp_A3} - H_{Ho}^{Hcp_A3} \quad (3)$$

$$\mathrm{GHSER_{Ga}} = {}^0G_{Ga}^{Orth} = G_{Ga}^{Orth} - H_{Ga}^{Orth}. \quad (4)$$

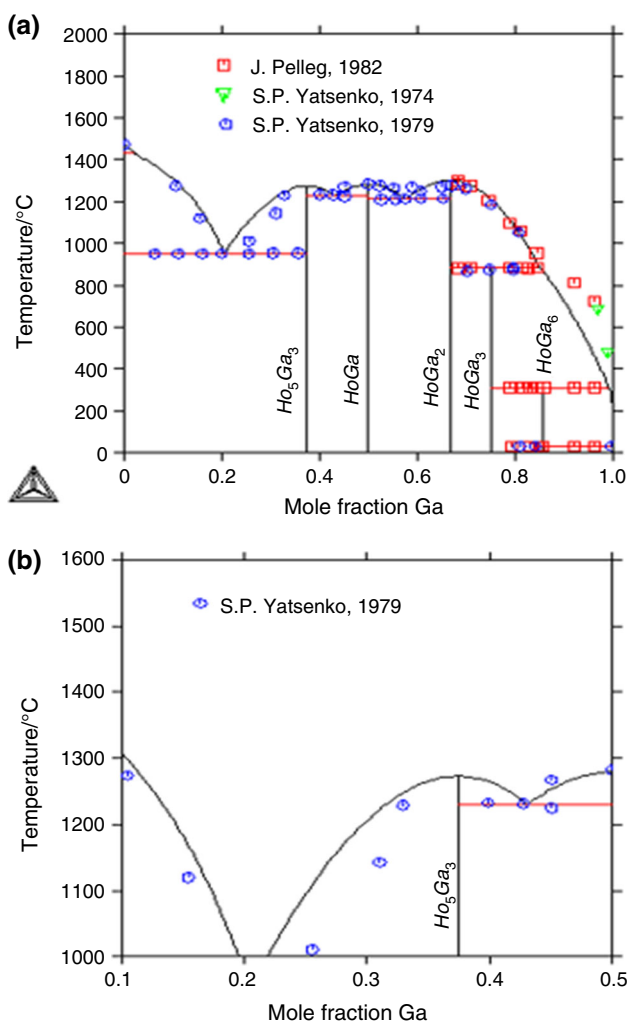

Fig. 3 **a** Calculated Ho–Ga system (Redlich–Kister) compared with the experimental data by [3, 4, 9], **b** enlarged part of Fig. 3

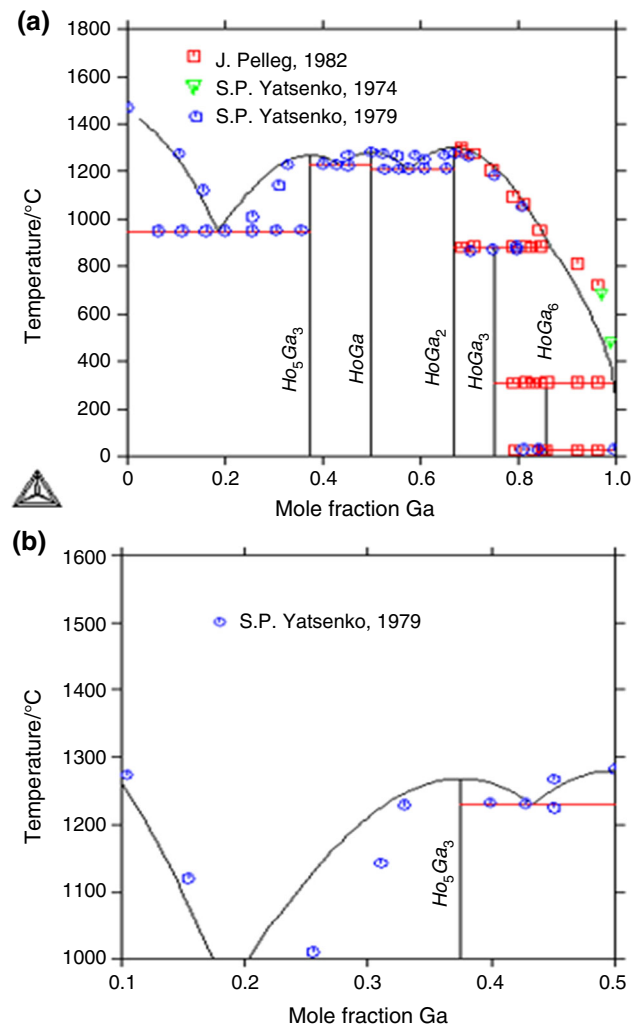

Fig. 4 **a** Calculated Ho–Ga system (Kaptay equation) compared with the experimental data by [3, 4, 9], **b** enlarged part of Fig. 4

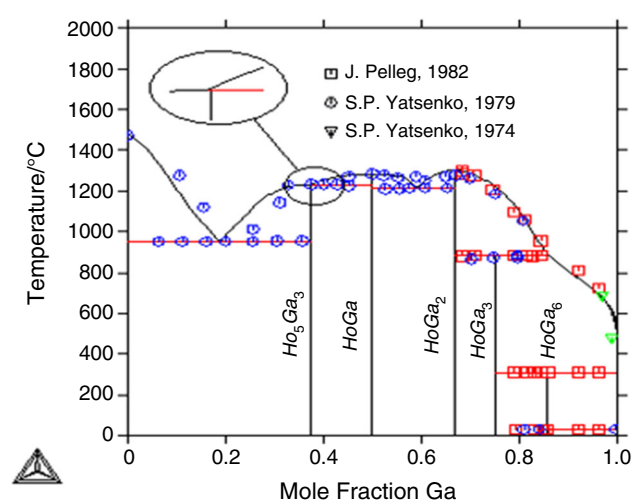

Fig. 5 Calculated Ho–Ga phase diagram using the association solution model compared with the experimental data by [3, 4, 9]

Table 2 Crystal data of the intermetallic phases in the Ho–Ga system

Diagram symbol	Composition at.% Ga	Crystal system	Structure type	References
αHo	0	Hex	–	–
Ho_5Ga_3	37.5	Hex	Mn_5Si_3	[19]
HoGa	50.0	Orth	CrB	[20, 21]
Ho_3Ga_5	62.5	Orth	Ga_5Tm_3	[10]
$HoGa_2$	66.7	Hex	AlB_2	[22]
$HoGa_3$	75.0	Cub	$AuCu_3$	[13, 23]
$HoGa_6$	87.5	Tetr	$PuGa_6$	[8]
Ga	100.0	Orth	–	–

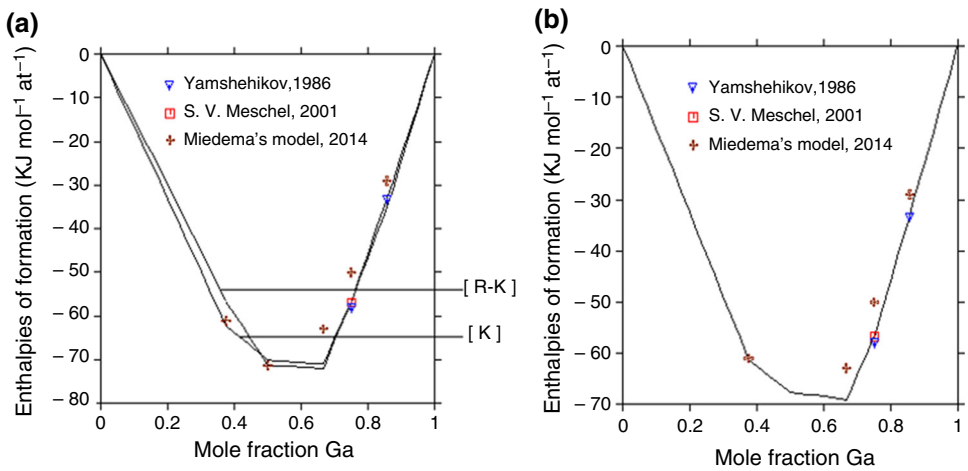

Fig. 6 **a** Calculated enthalpies of formation of the intermetallic compounds at 298.15 K using Redlich–Kister [R–K] and Kaptay [K] equations compared with the experimental data. **b** Calculated enthalpies of formation of the intermetallic compounds at 298.15 K using the association solution model compared with the experimental data

Thermodynamic description of solution phases

The calculation of the equilibrium between phases in a multi-constituted system requires the minimization of the total Gibbs free energy G of all the phases that exist in this equilibrium:

$$G = \sum_{i=1}^{n_p} n_i G_i^{\varphi} = \text{minimum}. \quad (5)$$

where n_p represents the total number of phases of the system, n_i is the number of moles and G_i^{φ} is the partial free Gibbs energy of the constituent i in the phase φ.

In the Ho-Ga system, the solution phases (Hcp, orthorhombic and liquid) have been modelled at first as substitution solutions using the Redlich–Kister polynomial [25]. The molar Gibbs energy of the φ phase is expressed by the sum of the reference part ${}^{ref}G^{\varphi}$, the ideal part ${}^{id}G^{\varphi}$ and the excess part ${}^{exc}G^{\varphi}$:

$$G_m^{\varphi} = {}^{ref}G^{\varphi} + {}^{id}G^{\varphi} + {}^{exc}G^{\varphi}. \quad (6)$$

As used in Thermo-Calc software [26]:

$${}^{ref}G^{\varphi}(T) = ({}^0G_{Ho}^{\varphi}(T) - H_{Ho}^{SER})x_{Ho} + ({}^0G_{Ga}^{\varphi}(T) - H_{Ga}^{SER})x_{Ga} \quad (7)$$

$${}^{id}G^{\varphi} = RT\sum_i x_i \ln x_i = RT(x_{Ho}\ln x_{Ho} + x_{Ga}\ln x_{Ga}) \quad (8)$$

where R is the gas constant, T is the temperature, in Kelvin (K), and x_{Ho} and x_{Ga} are the mole fraction of the elements Ho and Ga, respectively.

The excess Gibbs energy ${}^{exc}G_m^{\varphi}(T)$ of the φ phase is modelled by the Redlich–Kister polynomial [25] formula as follows:

$${}^{exc}G_m^{\varphi}(T) = x_{Ho}x_{Ga}\sum_{i=0}^{n} {}^iL_{Ho,\,Ga}^{\varphi}(T)(x_{Ho} - x_{Ga})^i \quad (9)$$

where ${}^iL_{Ho,\,Ga}^{\varphi}(T)$ is the binary interaction parameter of the two constituents Ho and Ga in the solution φ, which is to be evaluated in the present work. Its general form is:

$${}^iL_{Ho,Ga}^{\varphi}(T) = a_i + b_iT \quad (i = 0, 1, 2\ldots). \quad (10)$$

where a_i and b_i are the coefficients to be optimized and $b_i = -s$ is the excess entropy.

Chen et al. [27] reported that during the procedure of optimizing phase diagrams by linear dependence, for some high-temperature values the term excess energy changes sign; therefore, the constraint $d^2G/dx^2 > 0$ will change of sign. Hence, the appearance of the miscibility gap reversed a phase φ. To solve this problem, Kaptay [28–30] proposes the following semi-empirical equation:

Table 3 Comparison between experimental and calculated data (using Redlich–Kister, Kaptay equations and association solution model) relative to the Ho–Ga invariant reactions

Invariant reactions	This work						References [9]		References [4]	
	(Redlich equation)		(Kaptay equation)		(Association model)					
	x (Ga)/at	T/°C	x (Ga)/at	T/°C	x (Ga)/at	T/°C	x (Ga)/at	T/°C	x (Ga)/at	T/°C
Liq ⇔ Ho_5Ga_3 + (Ho)	0.204	950	0.183	950	0.190	950	–	–	0.202	950
Liq ⇔ Ho_5Ga_3 + HoGa	0.430	1230	0.433	1230	0.375	1230	–	–	0.375	1230
Liq ⇔ HoGa	0.5	1280	0.5	1280	0.5	1280	–	–	0.5	1280
Liq ⇔ $HoGa_2$ + HoGa	0.576	1210	0.576	1210	0.586	1210	–	–	0.575	1210
Liq ⇔ $HoGa_2$	0.667	1300	0.667	1300	0.667	1300	0.667	1300	0.667	1270
$HoGa_3$ ⇔ Liq + $HoGa_2$	0.85	885	0.86	885	0.87	885	0.857	885 ± 5	0.866	870
$HoGa_6$ ⇔ Liq + $HoGa_3$	0.991	312	0.996	312	1.00	312	1.000	312 ± 7	–	–
(Ga) ⇔ Liq + $HoGa_6$	0.999	28	0.999	26	1.00	28	1.000	28 ± 2	–	30

Table 4 Calculated thermodynamic parameters of the holmium–gallium system using Redlich–Kister equation [25]

Phases	Thermodynamic models	Parameters
Liquid	$(\mathrm{Ho},\ \mathrm{Ga})_1$	${}^0L^{\mathrm{liq}} = -211943.735 + 22.8215820 * T$
		${}^1L^{\mathrm{liq}} = -5313.57407 - 29.5624427 * T$
		${}^2L^{\mathrm{liq}} = 25178.4331 - 10.2337078 * T$
HCP_A3	$(\mathrm{Ho},\ \mathrm{Ga})_1(\mathrm{Va})_{.5}$	No excess term optimized
ORT	(Ga)	No excess term optimized
Ho_5Ga_3	$(\mathrm{Ho})_{0.625}$: $(\mathrm{Ga})_{0.375}$	$G^{\mathrm{Ho_5Ga_3}}_{\mathrm{Ho,\ Ga}} - 0.625^{298}\mathrm{H}^{\mathrm{HCP_A3}}_{\mathrm{Ho}} - 0.375^{298}\mathrm{H}^{\mathrm{ORT}}_{\mathrm{Ga}}$
		$= 0.625G^{\mathrm{HCP_A3}}_{\mathrm{Ho}} + 0.375G^{\mathrm{ORT}}_{\mathrm{Ga}} - 56,457.3896 + 1.34933937 * \mathrm{T}$
HoGa	$(\mathrm{Ho})_{0.500}$: $(\mathrm{Ga})_{0.500}$	$G^{\mathrm{HoGa}}_{\mathrm{Ho,\ Ga}} - 0.500^{298}H^{\mathrm{HCP_A3}}_{\mathrm{Ho}} - 0.500^{298}H^{\mathrm{ORT}}_{\mathrm{Ga}}$
		$= 0.500G^{\mathrm{HCP_A3}}_{\mathrm{Ho}} + 0.500G^{\mathrm{ORT}}_{\mathrm{Ga}} - 71,200 + 4.71162411 * T$
$HoGa_2$	$(\mathrm{Ho})_{0.333}$: $(\mathrm{Ga})_{0667}$	$G^{\mathrm{HoGa_2}}_{\mathrm{Ho,Ga}} - 0.333^{298}H^{\mathrm{HCP_A3}}_{\mathrm{Ho}} - 0.667^{298}H^{\mathrm{ORT}}_{\mathrm{Ga}}$
		$= 0.333G^{\mathrm{HCP_A3}}_{\mathrm{Ho}} + 0.667G^{\mathrm{ORT}}_{\mathrm{Ga}} - 72,000 + 3.59070837 * T$
$HoGa_3$	$(\mathrm{Ho})_{0.250}$: $(\mathrm{Ga})_{0.750}$	$G^{\mathrm{HoGa_3}}_{\mathrm{Ho,Ga}} - 0.250^{298}H^{\mathrm{HCP_A3}}_{\mathrm{Ho}} - 0.750^{298}H^{\mathrm{ORT}}_{\mathrm{Ga}}$
		$= 0.250G^{\mathrm{HCP_A3}}_{\mathrm{Ho}} + 0.750G^{\mathrm{ORT}}_{\mathrm{Ga}} - 56,800 + 0.1140342212 * T$
$HoGa_6$	$(\mathrm{Ho})_{0.143}$: $(\mathrm{Ga})_{0.857}$	$G^{\mathrm{HoGa_6}}_{\mathrm{Ho,Ga}} - 0.143^{298}H^{\mathrm{HCP_A3}}_{\mathrm{Ho}} - 0.857^{298}H^{\mathrm{ORT}}_{\mathrm{Ga}}$
		$= 0.143G^{\mathrm{HCP_A3}}_{\mathrm{Ho}} + 0.857G^{\mathrm{ORT}}_{\mathrm{Ga}} - 35,000 + 0.834202118 * T$

$$L^{\varphi}_{\mathrm{i,\ j}} = H_{\mathrm{i}} \exp\left(-\frac{T}{\tau_{\mathrm{i}}}\right) \tag{11}$$

with $\tau_{\mathrm{i}} = \frac{H_{\mathrm{i}}}{S_{\mathrm{i}}}$.

Where H_{i}. is the enthalpy part and S_{i} the excess entropy of $L^{\mathrm{Liquid}}_{\mathrm{Ho,\ Ga}}$ at $T = 0$ K; in general H_{i} and S_{i} have the same sign; therefore, τ_{i} is often positive. Equation 11 becomes Eq. (12) with the i = Ho, j = Ga and φ = Liquid.

$$L^{\mathrm{Liquid}}_{\mathrm{Ho,Ga}} = H_{\mathrm{i}} \exp\left(\left(-\frac{S_{\mathrm{i}}}{H_{\mathrm{i}}}\right) * T\right) \tag{12}$$

where the parameters H_{i} and S_{i} are to be optimized in the current work.

Associate model

When the short-range order is present in a disordered solution at a great distance, especially in a metallic liquid, we can imagine the formation of stoichiometric complexes in the solution. These complexes, or associated species, can be a simple mathematical instrument of modelling or represent real chemically stable entities. Sommer [31] developed this model.

The development of an associated model consists in defining the stoichiometry of the 'molecules' present in the liquid. In our case, '$HoGa_2$' is the compound with the highest melting and which solidifies congruently; we thought it was reasonable to assume that Ho–Ga liquids

Table 5 Optimized thermodynamic parameters of the holmium–gallium system using Kaptay equation [28–30]

Phases	Thermodynamic models	Parameters
Liquid	$(\mathrm{Ho, Ga})_1$	${}^0L^{\mathrm{liq}} = -195540.711 * \mathrm{Exp}(-3.23391319E-07 * T)$
		${}^1L^{\mathrm{liq}} = -24617.4423 * \mathrm{Exp}(-6.67906463E-05 * T)$
		${}^2L^{\mathrm{liq}} = 22739.1025$
HCP_A3	$(\mathrm{Ho,\ Ga})_1(\mathrm{Va})_{.5}$	No excess term optimized
ORT	(Ga)	No excess term optimized
Ho_5Ga_3	$(\mathrm{Ho})_{0.625}$: $(\mathrm{Ga})_{0.375}$	$G^{\mathrm{Ho_5Ga_3}}_{\mathrm{Ho,Ga}} - 0.625^{298}H^{\mathrm{HCP_A3}}_{\mathrm{Ho}} - 0.375^{298}H^{\mathrm{ORT}}_{\mathrm{Ga}}$
		$= 0.625G^{\mathrm{HCP_A3}}_{\mathrm{Ho}} + 0.375G^{\mathrm{ORT}}_{\mathrm{Ga}} - 62,219.7971 + 1.32386994 * T$
HoGa	$(\mathrm{Ho})_{0.500}$: $(\mathrm{Ga})_{0.500}$	$G^{\mathrm{HoGa}}_{\mathrm{Ho,Ga}} - 0.500^{298}H^{\mathrm{HCP_A3}}_{\mathrm{Ho}} - 0.500^{298}H^{\mathrm{ORT}}_{\mathrm{Ga}}$
		$= 0.500G^{\mathrm{HCP_A3}}_{\mathrm{Ho}} + 0.500G^{\mathrm{ORT}}_{\mathrm{Ga}} - 70,000 + 0,889633234 * T$
$HoGa_2$	$(\mathrm{Ho})_{0.333}$: $(\mathrm{Ga})_{0667}$	$G^{\mathrm{HoGa_2}}_{\mathrm{Ho,Ga}} - 0.333^{298}H^{\mathrm{HCP_A3}}_{\mathrm{Ho}} - 0.667^{298}H^{\mathrm{ORT}}_{\mathrm{Ga}}$
		$= 0.333G^{\mathrm{HCP_A3}}_{\mathrm{Ho}} + 0.667G^{\mathrm{ORT}}_{\mathrm{Ga}} - 71,000 + 1.82999138 * T$
$HoGa_3$	$(\mathrm{Ho})_{0.250}$: $(\mathrm{Ga})_{0.750}$	$G^{\mathrm{HoGa_3}}_{\mathrm{Ho,Ga}} - 0.250^{298}H^{\mathrm{HCP_A3}}_{\mathrm{Ho}} - 0.750^{298}H^{\mathrm{ORT}}_{\mathrm{Ga}}$
		$= 0.250G^{\mathrm{HCP_A3}}_{\mathrm{Ho}} + 0.750G^{\mathrm{ORT}}_{\mathrm{Ga}} - 56,800 - 0.0523417938 * T$
$HoGa_6$	$(\mathrm{Ho})_{0.143}$: $(\mathrm{Ga})_{0.857}$	$G^{\mathrm{HoGa_6}}_{\mathrm{Ho,Ga}} - 0.143^{298}H^{\mathrm{HCP_A3}}_{\mathrm{Ho}} - 0.857^{298}H^{\mathrm{ORT}}_{\mathrm{Ga}}$
		$= 0.143G^{\mathrm{HCP_A3}}_{\mathrm{Ho}} + 0.857G^{\mathrm{ORT}}_{\mathrm{Ga}} - 33,000 - 2.66853243 * T$

Table 6 Calculated thermodynamic parameters of the holmium–gallium system using the association solution model [31]

Phases	Thermodynamic models	Parameters
Liquid	$(\mathrm{Ho,\ Ga,\ HoGa_2})_1$	${}^0G^{\mathrm{Liq}} - {}^{298}\mathrm{H}^{\mathrm{Hcp_A3}}_{\mathrm{Ho}} - {}^{298}\mathrm{H}^{\mathrm{Orth}}_{\mathrm{Ga}} = -128662.358 + 19.5934221 * T$
		${}^0L^{\mathrm{Liq}}_{\mathrm{Ho,Ga}} = 62504.4545 - 81.7263970 * T$
		${}^0L^{\mathrm{Liq}}_{\mathrm{Ga,HoGa_2}} = 21331.3025 - 79.5841381 * T$
		${}^0L^{\mathrm{Liq}}_{\mathrm{HoGa_2,Ho}} = -292563.581 + 113.215848 * T$
		${}^1L^{\mathrm{Liq}}_{\mathrm{HoGa_2,Ho}} = -109393.807 + 39.2855319 * T$
		${}^2L^{\mathrm{Liq}}_{\mathrm{HoGa_2,Ho}} = -205736.050 - 242.024274 * T$
HCP_A3	$(\mathrm{Ho,\ Ga})_1(\mathrm{Va})_{.5}$	No excess term optimized
ORT	(Ga)	No excess term optimized
Ho_5Ga_3	$(\mathrm{Ho})_{0.625}$: $(\mathrm{Ga})_{0.375}$	$G^{\mathrm{Ho_5Ga_3}}_{\mathrm{Ho,\ Ga}} - 0.625^{298}\mathrm{H}^{\mathrm{HCP_A3}}_{\mathrm{Ho}} - 0.375^{298}\mathrm{H}^{\mathrm{ORT}}_{\mathrm{Ga}}$
		$= 0.625G^{\mathrm{HCP_A3}}_{\mathrm{Ho}} + 0.375G^{\mathrm{ORT}}_{\mathrm{Ga}} - 61,200 + 9 * \mathrm{T}$
HoGa	$(\mathrm{Ho})_{0.500}$: $(\mathrm{Ga})_{0.500}$	$G^{\mathrm{HoGa}}_{\mathrm{Ho,\ Ga}} - 0.500^{298}H^{\mathrm{HCP_A3}}_{\mathrm{Ho}} - 0.500^{298}H^{\mathrm{ORT}}_{\mathrm{Ga}}$
		$= 0.500G^{\mathrm{HCP_A3}}_{\mathrm{Ho}} + 0.500G^{\mathrm{ORT}}_{\mathrm{Ga}} - 67,600 + 7.33088545 * T$
$HoGa_2$	$(\mathrm{Ho})_{0.333}$: $(\mathrm{Ga})_{0667}$	$G^{\mathrm{HoGa_2}}_{\mathrm{Ho,Ga}} - 0.333^{298}H^{\mathrm{HCP_A3}}_{\mathrm{Ho}} - 0.667^{298}H^{\mathrm{ORT}}_{\mathrm{Ga}}$
		$= 0.333G^{\mathrm{HCP_A3}}_{\mathrm{Ho}} + 0.667G^{\mathrm{ORT}}_{\mathrm{Ga}} - 69,046.2765 + 3.80115934 * T$
$HoGa_3$	$(\mathrm{Ho})_{0.250}$: $(\mathrm{Ga})_{0.750}$	$G^{\mathrm{HoGa_3}}_{\mathrm{Ho,Ga}} - 0.250^{298}H^{\mathrm{HCP_A3}}_{\mathrm{Ho}} - 0.750^{298}H^{\mathrm{ORT}}_{\mathrm{Ga}}$
		$= 0.250G^{\mathrm{HCP_A3}}_{\mathrm{Ho}} + 0.750G^{\mathrm{ORT}}_{\mathrm{Ga}} - 56,800 + 2.68982686 * T$
$HoGa_6$	$(\mathrm{Ho})_{0.143}$: $(\mathrm{Ga})_{0.857}$	$G^{\mathrm{HoGa_6}}_{\mathrm{Ho,Ga}} - 0.143^{298}H^{\mathrm{HCP_A3}}_{\mathrm{Ho}} - 0.857^{298}H^{\mathrm{ORT}}_{\mathrm{Ga}}$
		$= 0.143G^{\mathrm{HCP_A3}}_{\mathrm{Ho}} + 0.857G^{\mathrm{ORT}}_{\mathrm{Ga}} - 33,100 - 1.26835082 * T$

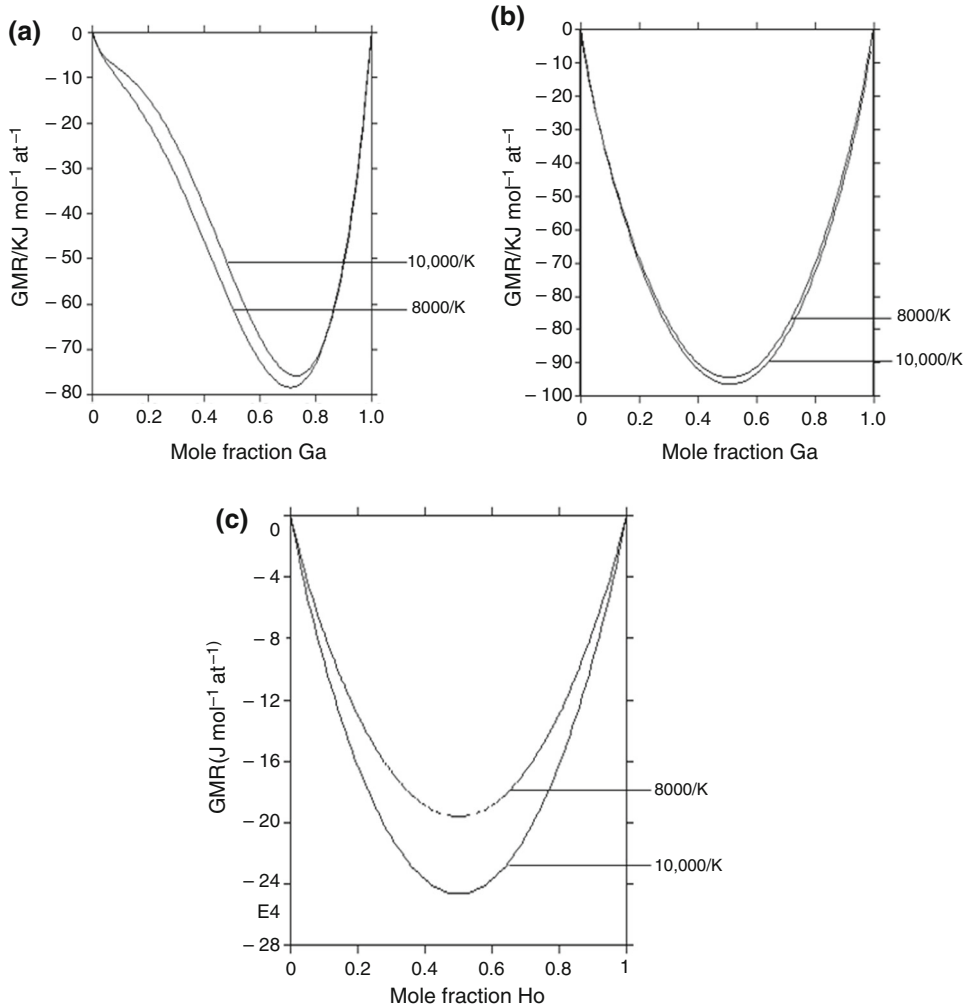

Fig. 7 **a** Calculated Gibbs free energy of mixing of the liquid phase at different temperatures (8000–100,00 K) using the Redlich–Kister model [25]. **b** Calculated Gibbs free energy of mixing of the liquid phase at different temperatures (8000–10,000 K) using the Kaptay model [28–30]. **c** Calculated Gibbs free energy of mixing of the liquid phase at different temperatures (8000–10,000 K) using the association solution model [31]

would tend to order around these compositions. The species present should therefore be Ho, Ga and $HoGa_2$. The resulting expression for the Gibbs energy is:

$$G_m^L = \sum_{i=Ho,\,Ga,\,HoGa_2} y_i^{\varphi 0} G_i^L + RT \times \sum_{i=Ho,\,Ga,\,HoGa_2} y_i^L \ln\left(y_i^L\right) + {}^{exc}G_m^L. \tag{13}$$

The Gibbs energy for the $HoGa_2$ associate is given by the following formula:

$${}^0G_{HoGa_2}^L = {}^0\Delta_f H_{HoGa_2}^L - T\,{}^0\Delta_f S_{HoGa_2}^L + {}^0G_{Ho}^L + 2\,{}^0G_{Ga}^L. \tag{14}$$

The excess part of the Gibbs energy of the liquid phase could be completely described as a result of interactions between Ho and $HoGa_2$ by the expression below:

$${}^{exc}G_m^L = y_{Ga} y_{HoGa_2} \sum_k {}^kL_{Ga,\,HoGa_2}^L \left(y_{HoGa_2}^L - y_{Ga}^L\right)^k. \tag{15}$$

According to this model, if the molar fraction of the hypothetical $HoGa_2$ associate were to decrease with temperature, for example, the ${}^{exe}G$ of Gibbs energy would also decrease, making the solution ideal and eliminating the problem related to the formation of miscibility gaps reversed at high temperature, and also solve the asymmetry located in the 50–70 at.% Ga range.

Thermodynamic description of stoichiometric compounds

In the Ho–Ga binary system, all intermediate compounds, Ho_5Ga_3, HoGa, $HoGa_2$, $HoGa_3$, and $HoGa_6$ are treated as stoichiometric phases and described as Ho_pGa_q.

The Gibbs free energy of the stoichiometric compounds Ho_pGa_q is expressed as follows:

$${}^0G_{Ho_pGa_q} = \frac{p}{p+q}\,{}^0G_{Ho} + \frac{q}{p+q}\,{}^0G_{Ga} + a + bT \tag{16}$$

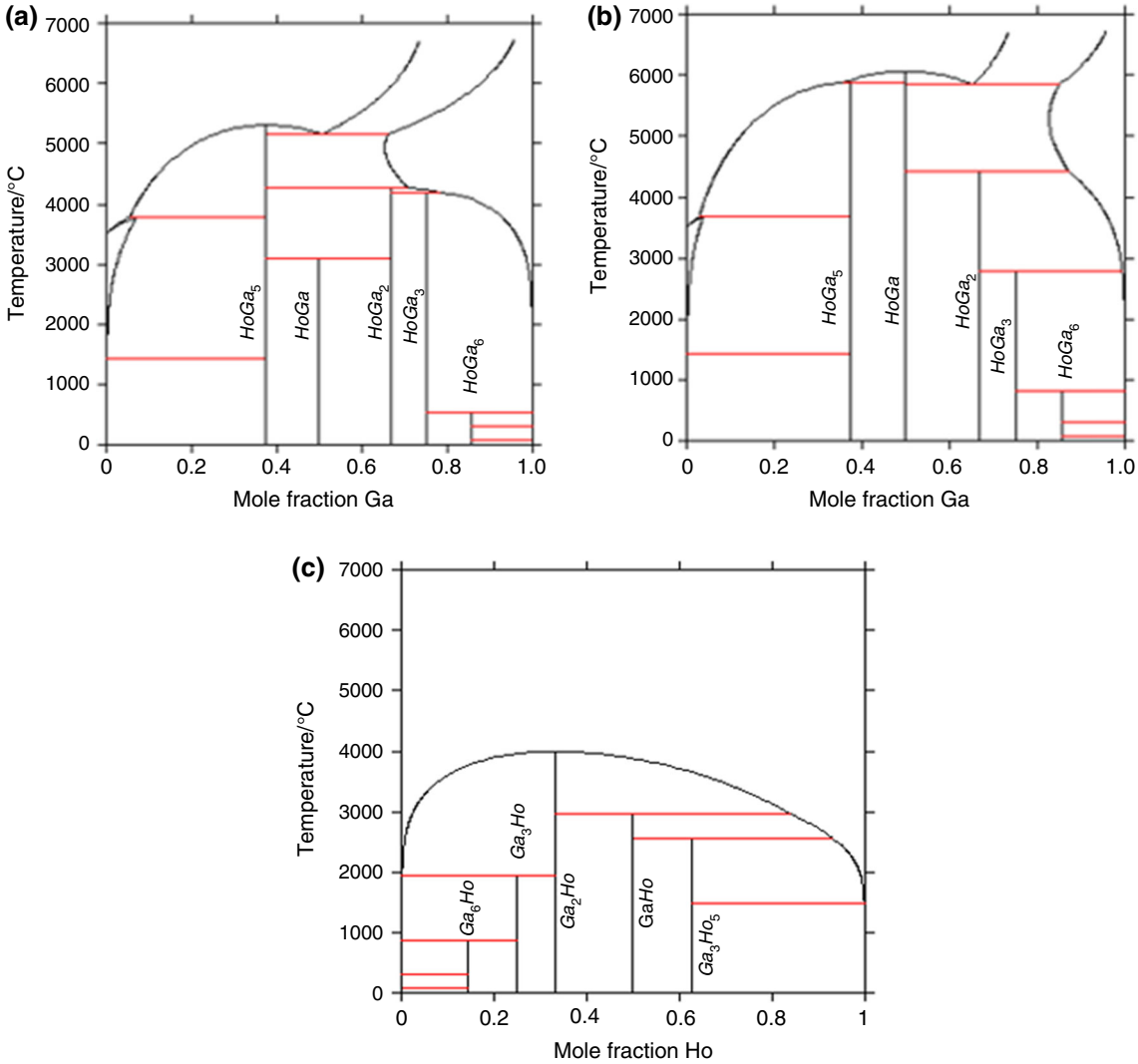

Fig. 8 **a** Calculated Ho–Ga phase diagram with the Redlich–Kister equation [25] when the liquid phase is suspended. **b** Calculated Ho–Ga phase diagram with the Kaptay equation [28–30] when the liquid phase is suspended. **c** Calculated Ho–Ga phase diagram with the association solution model [31] when the liquid phase is suspended

where ${}^{0}G_{\mathrm{Ho}}$ and ${}^{0}G_{\mathrm{Ga}}$ are the Gibbs energy of the pure elements Ho and Ga, respectively, and a and b are parameters to be determined for each of the compounds of the Ho − Ga system.

Results and discussion

The liquid phase has been described by three models: the Redlich–Kister model, the Kaptay model and the association solution model with a '$HoGa_2$' associate. Only the five intermetallic phases:Ho_5Ga_3,HoGa, $HoGa_2$, $HoGa_3$ and $HoGa_6$ were taken into account and treated as stoichiometric phases.

Using a set of thermodynamic parameters, the calculated phase diagram of the holmium–gallium system is presented in Figs. 3–5. In Fig. 3, we compared our results obtained by the Redlich–Kister model [25] with the data found in the literature [3, 4] and [9]. In Fig. 4, our phase diagram calculated by the Kaptay model [28–30] is compared with the same results of the literature. In Fig. 5, our phase diagram calculated using the association solution model [31] is compared with the same results of the literature.

We noted that our calculated values for the intermetallic compounds are in good agreement with the compositions and temperatures found experimentally by Yatsenko et al. [4] and Pelleg and Kimmel [9], especially the results found by the association solution model. In the case of optimization with the Redlich–Kister method [25], the slope of the liquid phase decreases from the peritectic point of the $HoGa_6$ phase to T = 312 °C more stably than in the case of optimization with the Kaptay formula [28–30] for which a

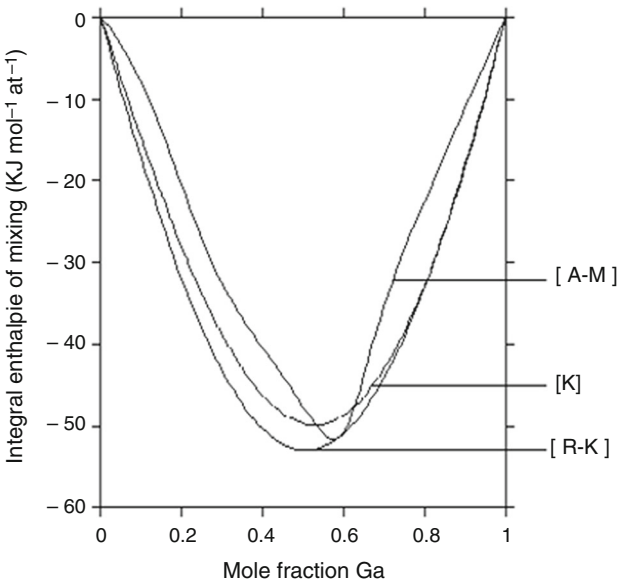

Fig. 9 Calculated integral enthalpies of formation of the liquid phase at T = 1600 K, using the Redlich–Kister Model [R–K], Kaptay model [K] and the association solution model [A-M]

higher slope is observed between the temperatures 312 °C and 855 °C. But for the association solution model [31], the liquid phase is more constant and very similar to that of the experimental for all points of the system.

The diagrams calculated using the Redlich–Kister and Kaptay methods are identical, and the diagram calculated by the association solution model is very similar to the experimental diagram of [4] and [9]. However, an exception noted for the phase Ho_5Ga_3 is a little detached, which is formed by the following peritectic reaction HoGa ⇔ Liq + Ho_5Ga_3 (0.375 at.% Ga) in [4, 9]. The liquid phase and the intermetallic compound Ho_5Ga_3 have the same composition. To solve the problem of the anomalous asymmetry of liquidus between 50 and 70 at.% Ga, our optimizations by the Redlich–Kister model and Kaptay model guided to a congruent point, for the compound Ho_5Ga_3 at 1260 °C and 0.375 at.% Ga (Redlich–Kister)/at 1270 °C and 0.375 at.% Ga (Kaptay), and a eutectic reaction Liq ⇔ HoGa + Ho_5Ga_3 at 1230 °C and 0.430 at.% Ga (Figs. 3, 4) instead of a peritectic reaction. The difference in composition between the liquid phase and the compound Ho_5Ga_3 is very low from 0.05 at.% Ga. We tried to minimize this difference and obtain a better description of the liquid phase in the holmium-rich side, introducing the model of association solution through which the liquid phase was perfectly treated and the problem of the asymmetry of the experimental is solved. The three phase diagrams obtained in the current work using the three models are identical, as previously mentioned. For the comparison of the composition and temperature calculations of the invariant equilibria for this phase diagram using the three models, the results found are almost identical, and they are presented in Table 3.

In addition, our description of the calculated enthalpies and entropies of formation of the intermediate phases of the Ho–Ga binary system, involved in the present optimization, is compared with the experimental results of [16, 17]. The values predicted by the Miedema model [18] are illustrated in Fig. 6 and Table 1, as we can see that the enthalpy formation of the phase $HoGa_2$ is the most negative and that the phase $HoGa_6$ is the least negative. A satisfactory agreement is marked between the calculated values and the experimental values of the compounds $HoGa_3$ and $HoGa_6$ studied by [16, 17]. Consequently, the enthalpies of formation values predicted by the Miedema model [18] for the $HoGa_2$, HoGa and Ho_5Ga_3 phases are consistent with our optimized results. The enthalpy of formation part is identical in the three models, with the exception of the phase Ho_5Ga_3, which is more negative for the Kaptay formalism compared to the other two models, and the entropic part is almost identical for the Kaptay and Redlich–Kister models, but a bit high for the association solution model. Tables 4, 5 and 6 present the thermodynamic parameters optimized for all phases of the Ho–Ga system obtained using three formalisms.

In Fig. 7a–c, the Gibbs energy of mixing of the liquid phase is calculated with the three models at different temperatures: 8000–10000 K for the Ho–Ga system. In Fig. 7a (the Redlich–Kister model), the formation of an inverted miscibility gap is detected at very high temperatures. In Fig. 7b (Kaptay model) and Fig. 7c (association solution model), the liquid is stable at very high temperatures, no spinodal point was detected until a temperature of 10,000 K, and so no formation of an inverted miscibility gap has been detected.

In order to test the coherence of our thermodynamically calculated parameters, we have drawn the phase diagram of the Ho–Ga binary system when the liquid phase is suspended as indicated by the authors [32], the stoichiometric phases disappear at high temperature (5800 °C 5800°C) (Fig. 8a, b) for the Redlich–Kister and Kaptay formalisms, and however, for the association solution model, the stoichiometric phases disappear at 4000 °C (Fig. 8c). Terminal solid solutions are modelled. But in our case, it will be noted that at high temperature for the model of Redlich–Kister and Kaptay, only the solid solution Hcp is calculated over the entire range of gallium composition and not in the rich region, in the orthorhombic structure. The integral enthalpies of formation of the liquid phase calculated in this work over the whole range of composition at T = 1600 K, using the Redlich–Kister Model [R–K], Kaptay model [K] and the association solution model [A-M] is presented in Fig. 9; owing to the lack of experimental determinations, no comparison with the literature is possible.

Conclusions

The complete thermodynamic modelling of phase equilibria and experimental thermodynamic data for the holmium–gallium binary system available in the literature were performed using the CALPHAD method in which the Gibbs energy equations of the liquid phase were described by the Redlich–Kister polynomial, Kaptay and the association solution model.

In the three descriptions, a set of thermodynamic parameters was obtained; the calculated phase diagram and the calculated values are equivalent and in good agreement thermodynamically with the data of the literature. Due to the lack of experimental information on the mixing enthalpies of the liquid phase and the experimental determination of the formation enthalpies of the compounds $HoGa_2$, HoGa and Ho_5Ga_3, thermodynamic modelling is still waiting for a more precise determination.

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